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Versuch: Fluorescence Correlation Spektroscopy

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Contents

1 Preparation **3**
1.1 Introduction . . . 3
1.2 Theoretical Background . . . 3
1.2.1 Confocal Microscope . . . 3
1.2.2 Fluorescence . . . 3
1.2.3 Brownian Motion . . . 3
1.2.4 Autocorrelation . . . 4
1.2.5 Triplet Photonics . . . 4
1.3 Experimental Setup . . . 4

2 Analysis **6**
2.1 Measurement of a Silicon EEPROM Chip . . . 6
2.2 Measurement of Fluorescent Beads to estimate the Microscope Resolution . . . 8
2.3 FCS Measurement of Freely Diffusing Nanoparticles . . . 10
2.3.1 Calibration with Atto655 . . . 10
2.3.2 Characterisation of Nanoparticles . . . 11

1 Preparation

1.1 Introduction

Fluorescence Correlation Spectroscopy (FCS) enables the obersvation of samples that are too smal for a traditional microscope. This is acchieved with the usage of a confocal microscope and the observation of fluorescence fluctuations as fluorescent mollecules diffuse through a microscopic region.

1.2 Theoretical Background

The theoretical background for this lab curse experiment is a summarisation from [1].

1.2.1 Confocal Microscope

The resolution of a classical light microscope is limited by the Abbe limit, which can be described with the following formula:

$$d = \frac{\lambda}{2 \cdot n \cdot \sin(\alpha)} = \frac{\lambda}{2 \cdot NA} \,. \tag{1.1}$$

The minimal resolvable distance d is determined by the wavelength of the used light λ, the refractive index of the surrounding medium n and the half angle of the light cone α. It can be also described with the Numerical Apperture, which is defined as: $NA = n \cdot \sin(\alpha)$.
An improvement of this microscope is acchieved by using a point illumination with a laser and a pinhole sized apperture to eliminate light outside the focal point of a lense. The sample is now scanned point by point and the image is reconstructed on a computer. With this technique, confocal microscopes acchieve a higher resolution than the classical light microscope because the point illumniation. To detect the light, a Photomultiplier Tube (PMT) or a Single Photon Avalanche Diode (SPAD) can be used as a detector.

1.2.2 Fluorescence

After a photon was bsorbed by stiumlated emission, the electron which captured the photon is transitioning from its ground state to an excited state. From this excited state the photon can return to its ground state by emitting a photon. In case of fluorescence this fall-back to the ground state happens shortly after the excitation. On the way to its ground state, the photon does not have to make the same transition backwards but it can first relax to the lowest vibrational level of the excited state before returning to the ground state. This phenomeon is called the "Stokes Shift" and it describes the differnce in the wavelength of the absorbed light and the emitted light, which has a higher wavelength.

1.2.3 Brownian Motion

Brownian motion describes the temperature depending random movement of particles. In Fluorescence Correlation Spectroscopy (FCS) it is the main contributer to the fluctuations in the intensity of the detected fluorescencing light. The Stokes-Einstein relation for diffusion describes the behaviour with the formula

$$D = \frac{k_B \cdot T}{6 \cdot \pi \cdot \eta \cdot r_0} \tag{1.2}$$

where D is the diffusion coefficient, k_B is the Boltzmann constant, T is the absolute temperature, η is the viscosity of the medium and r_0 is the hydrodynamic radius of the particle. In this experiment, the hydrodynamic radius r_0 is measured with FCS.

1.2.4 Autocorrelation

An autocorrelation is a convolution of a function with itself. For a function $F(t)$ the autocorrelation is defined as follows:

$$G(\tau) = \frac{\langle F(t)F(t+\tau)\rangle}{\langle F(t)\rangle^2} - 1\,. \tag{1.3}$$

The parameter τ is called the "lag time". In the case of FCS the autocorrelation function $G(\tau)$ decribes the temporal behaviour of fluorescence intensity of freely diffusing particles in a liquid solution with the following model:

$$G(\tau) = G(0) \cdot \left(1 + \frac{\tau}{\tau_D}\right)^{-1} \cdot \left(1 + \left(\frac{r_0}{z_0}\right)^2 \frac{\tau}{\tau_D}\right)^{-\frac{1}{2}}\,. \tag{1.4}$$

To simplify the equation, the characteristic time τ_D was introduced, which describes the time for which a molecule stays in the confocal volume. It is related to the diffusion coefficient D and the latera width r_0 under the following equation:

$$\tau_D = \frac{r_0^2}{4D}\,. \tag{1.5}$$

The Amplidute at zero lag time, $G(0)$, is the inverse of the average number of particles $\langle N\rangle$ with the steady-state concentration c in the observed volume V_{eff}:

$$G(0) = \frac{1}{\langle N\rangle} = \frac{1}{V_{eff}\cdot c} = \frac{1}{\pi^{\frac{3}{2}}\cdot r_0^2 \cdot z_0 \cdot c}\,. \tag{1.6}$$

The local steady-state concentration can be determined with a calibration measurement.

1.2.5 Triplet Photonics

Diffusion occurs on the illisecond timecsale, whiloe triplet state blinking, which is caused by intersystem crossings to a dark triplet sttate, occurs within microseconds. This results in a humo in the autocorrelation at the beginning of the measurement. The triplet contribution to the Autocorrelation can be described as follows:

$$G_{triplet} = 1 + T \cdot e^{-\frac{T}{\tau_T}} \tag{1.7}$$

where T ist the triplet state fraction and τ_T is the triplet relaxation time. This contribution will be visible in the measured autocorrelation function.

1.3 Experimental Setup

The microscope is driven by a HeNe as a light source at the wavelength of 632.8 nm. At the beginning of a Measurement the oscilloscope cna be used in an optical mode to adjust the focus. For FCS the scan unit is activated. The microscope then scans the probe with XY-galvo mirrors. Aftere the excitation the emitted light is collected by the microscope and guided to a beam splitter which is used to separate the laser from the fluorescent light. The fluorescent light is directed into a Photo Multiplyer Tube (PMT) or a Single Photon Avalanche Diode (SPAD) and measured as an electrical signal.
For different magnifications there are three different objectives available:

- An air objective with a magnification factor of 10 and $N_A = 0,25$
- An oil filled objective with a magnification of 60 and $N_A = 1,40$
- A water filled objective with a magnification of 63 and $N_A = 1,2$

The lab experiment consists of the following exercises:

1. To get familiar with the microscope, an silicon EPROM chip is placed under the air objective. The focus is adjusted until the chip can be seen with the eye. Afterwards, the laser is switched on an the silicon chip is scanned with the laser until a good image is captured. In this case, the fluorescent light is detected with the PMT.
2. A drop of oil with some 40 nm fluorescend beads is placed on top of the oil objective. With the laser and the SPAD some images of the beads are captured. The goal of this task is to get the FWHM values from some intensity profiles to calculate the resolution of the microscope.
3. The last task measures the diffusing of free molecules in water. It is done with the water objective and the SPAD detector. At first, Atto655 erference probe is analysed for calibration purposes. Afterwards the analysis is done with six different samples. The goal of this task is to determin the lateral width r_0 of the other samples.

2 Analysis

2.1 Measurement of a Silicon EEPROM Chip

In the first task a silicon EPROM Chip was to be measured with the use of the air objective. The objective has a magnification of 10x and a numerical aperture of 0.25. To place the chip on a propper position under the objective, the microscope was fist used in the regular mode to scan the probe with the eye. After finding a suitable position, the microscope was switched to FCS mode to scan the chip with the PMT. For further adjustments the PMT signal amplification was increased until the Signal to Noise Ration (SNR) was sufficient to see the structure of the chip. Then several pictures, each with a different pixel size, were taken. For each picture, two very close bright spots were chosen as maxima in order to determine the respective Rayleigh seperation using the intensity profile. The respective two maxima were different for each picture.

The results can be found in 2.1.

Table 2.1: Rayleigh seperation for scans with different pixel sizes

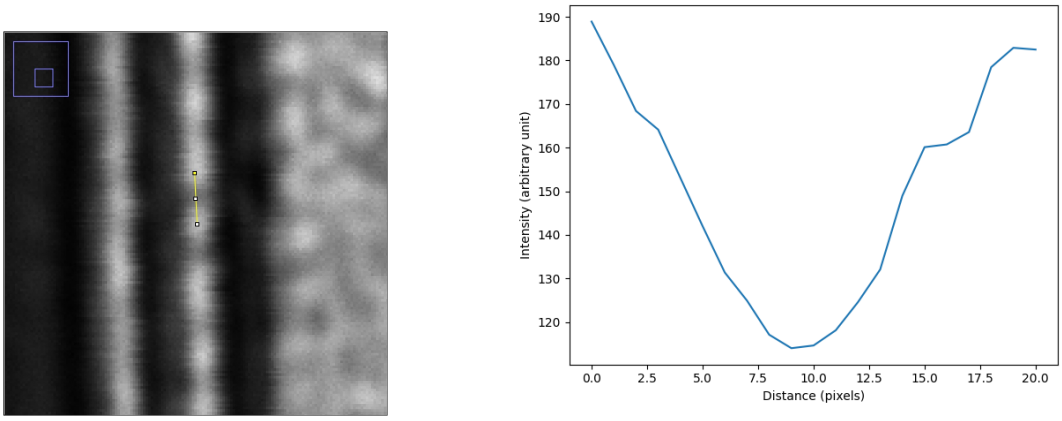
Pixel size in μm	0.1	0.2	0.3	0.4
Rayleigh seperation in px	21	13	8	6
Rayleigh seperation in μm	2.1	2.6	2.4	2.4

The theoretical value for the Rayleigh seperation is given by

$$d_{\min} = \frac{\lambda}{2\mathrm{NA}} = \frac{0.6328\,\mu m}{2 \cdot 0.25} \approx 1.27\,\mu m, \tag{2.1}$$

with the laser wavelength λ and the numerical aperture NA.

It is obvious that the experimental results are quite different from the theoretical value. A very propable reason is that the theory for the theoretical Rayleigh seperation assumes perfect



(a) Scan with line between two maxima (b) Intensity profile of line between two maxima

Figure 2.1: Scan and intensity profile (pixel size = $0.1\,\mu m$)

points for determining the Rayleigh seperation whereas in practice we are dealing with points of a finite size, so that their potential overlap plays a significant role.

2.2 Measurement of Fluorescent Beads to estimate the Microscope Resolution

For this task we installed the 60xOil immersion objective, on which we first placed a drop of oil, in the microscope. Then a microscope slide containing 40 nm fluorescent beads was placed on the microscope stage. With everything ready, a scan was performed by averaging 10 individual scans into on picture, so that the beads would be more visible. The pixel size was set to 50.1 nm.

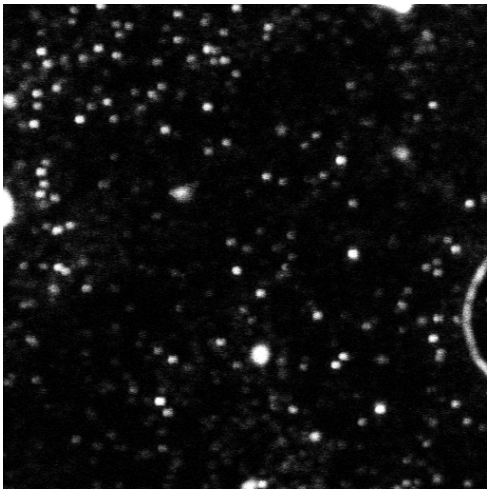


Figure 2.2: Scan of probe with nanobeads

For the first step of the analysis, ten pairs of close and clearly visible beads were chosen. For each of the pairs an analysis of the intensity profile between the beads and the corresponding full width half maximum (FWHM) values, ω_{FWHM}, were performed. The results can be seen in 2.2

ω_{FWHM} [px]	ω_{FWHM} [μm]
6.4817	0.3247
7.0716	0.3543
6.3216	0.3167
6.7446	0.3379
6.7699	0.3392
6.6894	0.3351
7.1165	0.3565
6.7216	0.3368
7.2355	0.3625
6.8275	0.3421

Table 2.2: ω_{FWHM} for 10 beads

Now, in order to determine the microscope resolution we use the relation

$$\omega_{\text{FWHM}}^2 = \omega_{\text{lat}}^2 + \omega_{\text{beads}}^2, \tag{2.2}$$

where ω_{lat} is the microscope resolution in μm and ω_{beads} is the bead size in μm.
Thus, the mean value and standard deviation from the values in 2.4 are:

$$\overline{\omega}_{\text{FWHM}} = 0.3406\,\mu m;\ \sigma_{\omega_{\text{FWHM}}} = 0.0141\,\mu m. \tag{2.3}$$

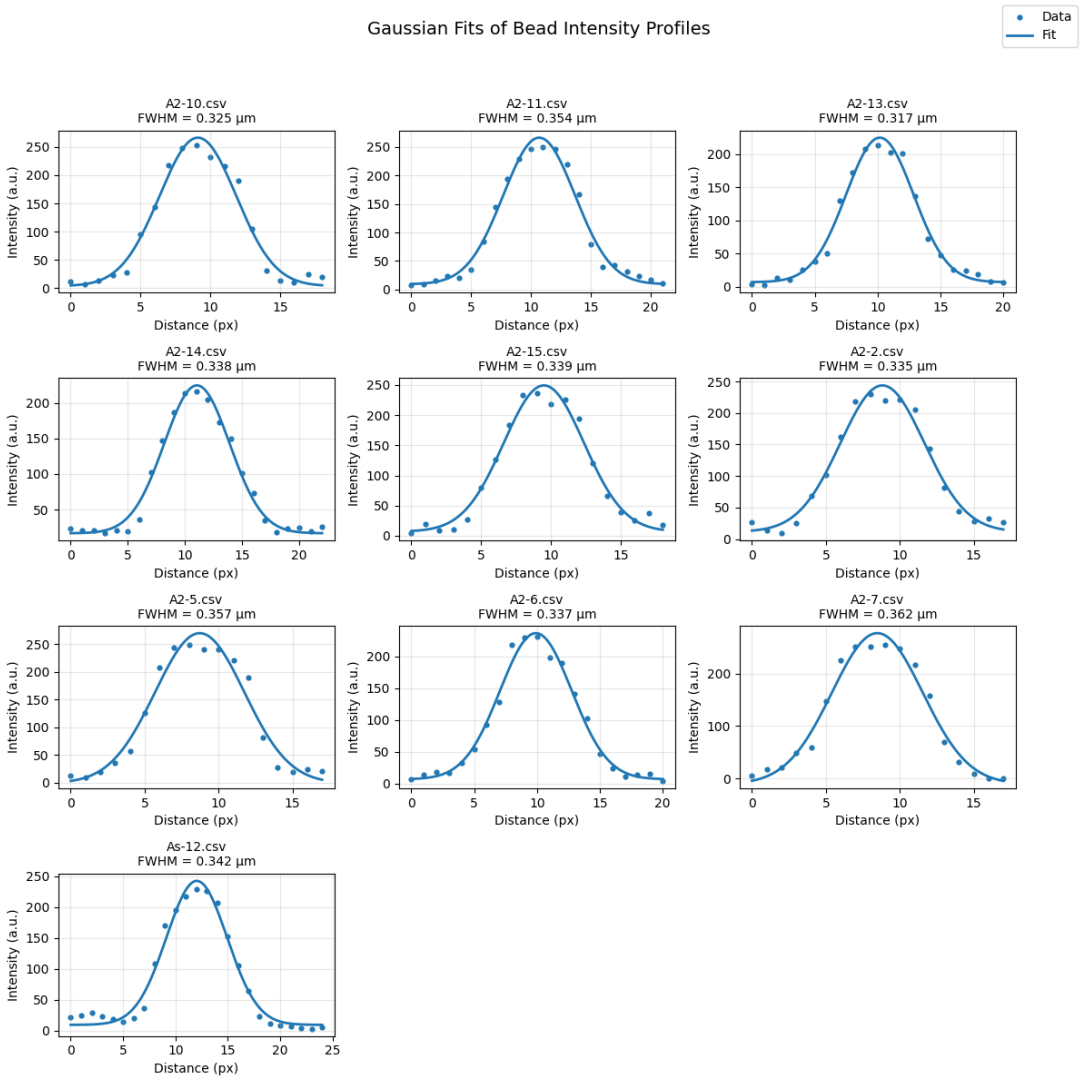


Figure 2.3: Fitted intensity profiles of 10 nanobeads

Thus, the corresponding value for the microscope (lateral) resolution is $\omega_{\text{lat}} = 0.3382\mu m$. The theretical value is given by:

$$\omega_{\text{lat, theo}} = 0.4 \cdot \frac{\lambda}{\text{NA}} = 0.4 \cdot \frac{0.6328\,\mu m}{1.4} \approx 0.181\,\mu m. \tag{2.4}$$

Here, our result differs strongly from the expected value, whereas the individual values for ω_{FWHM} are quite similar. This indicates, that the error is systemical.

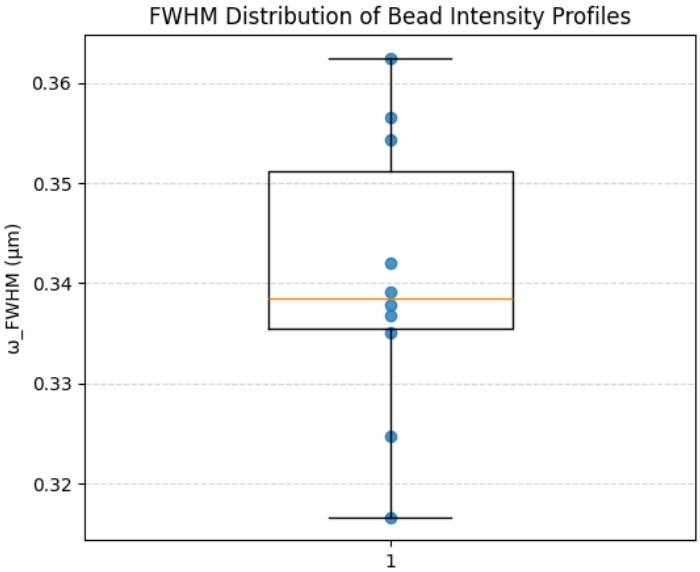


Figure 2.4: Distribution of FWHM values [μm]

2.3 FCS Measurement of Freely Diffusing Nanoparticles

The main goal of this task is the measurement of the hydrodynamic radius of nanoparticles of six unknown samples. Futhermore, the concentration of the samples can be detechted with this method. To utilise this method, the effective volume of focal volume must be determined with a reference sample of Affo655 with known diffusion coefficient. After this calibration the other samples can be analysed.

2.3.1 Calibration with Atto655

The autocorerlation function of the fluorescense intensity for the Atto655 sample, which is dissolved in water, is described with the model function

$$G(\tau) = G_0 \cdot \left(1 + \frac{\tau}{\tau_D}\right)^{-1} \cdot \left(1 + \frac{\tau}{\tau_D \cdot S^2}\right)^{-\frac{1}{2}} \tag{2.5}$$

which is described in 1.2.4. The influence of the triplet state is not taken into concideration at this model. Therefore, the fit will only be performed from 10 µs onwards in order to eliminate its influence on the fit. The value S is set to the constant $S = 5$ and the parameters G_0 and τ_D can be determined with a fit. The fit was performed during the data acquisition process with the results

$$G_0 = 0.0220(1) \tag{2.6}$$

$$\tau_D = 0.0659(1)\,\mathrm{ms}\,. \tag{2.7}$$

The measured data and the fit function are visualised in fig. 2.5.
The uncertainties are only reuslting from the fit. Other possibilities were not taken into account for the uncertainty estimation. With these parameters abd the diffusion coefficient of Atto655 the lateral width r_0 can be determined. The diffusion coefficient is temperature-dependant and with the measured room temperature of 20.0(5) °C and the konwn calculus from [1, Page 24] it is: 372.3(87) $\mu\mathrm{m}^2\,\mathrm{s}^{-1}$. With this, the lateral width results to:

$$r_0 = \sqrt{4 \cdot D \cdot \tau_D} = 0.313(5)\,\mu\mathrm{m}. \tag{2.8}$$

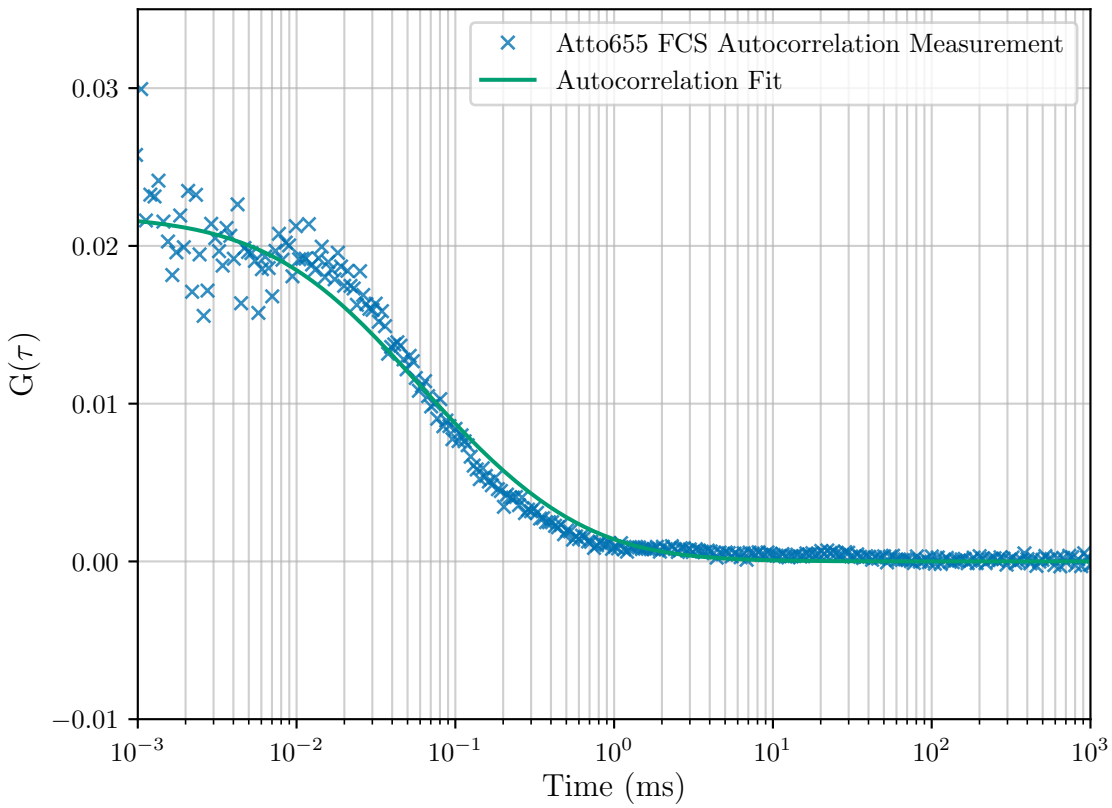


Figure 2.5: Experimental ACF data of diffusing Atto655 solved in water are shown as crosses. The solid line represent the fit according to eq. (2.5)

This value is only 64.8 % away from the literature value of 0.601 nm which can be found in [2, Page 46].
The FCS measurement was done with a water objective with a $N_A = 1.2$ whereas an oil objective with an $N_A = 1.4$ was used in the previous task. The resolution in water can be estimated with the Abbe formula for confocal microscopy:

$$\omega_{lat,water} \approx 0.4 \cdot \frac{\Lambda}{NA} \approx 210.9\,\mathrm{nm}\,. \tag{2.9}$$

The effective volume can be calculated with the following equation:

$$V_{eff} = \pi^{\frac{3}{2}} \cdot S \cdot r_0^3 = 0.858(43)\,\mathrm{fL}\,. \tag{2.10}$$

This value is useas a calibartion for the following analysis of different nanoparticles.

2.3.2 Characterisation of Nanoparticles

With the calibration from the previous task, six different nanoparticles are analysed to determine their hydrodynamic radius and concentrations. The FCS Autocorrelation measurments were already fitted durning the data acquisition. Table 2.3 lists the fit results for the different samples, including the amplitude at zero lag time G_0 and the diffusion time τ_D. A plot with all different autocorelation functions can be seen in fig. 2.6.
The hydrodynamic radius can be calculated with the following formula:

$$R_H = \frac{k_B \cdot T}{6 \cdot \pi \cdot \eta \cdot D} = \frac{4 \cdot k_B \cdot T \cdot \tau_D}{6 \cdot \pi \cdot \eta \cdot r_0^2} \tag{2.11}$$

Table 2.3: Analysis of the samples

Sample	G_0	τ_D in ms	R_H in nm	c in nM	$\langle N \rangle$
1	0.3445(3)	1.328(7)	11.6(4)	5.6(3)	2.903(3)
2	0.6813(8)	1.354(9)	11.8(4)	2.8(2)	1.468(2)
3	1.016(1)	1.293(8)	11.3(4)	1.9(1)	0.984(1)
4	0.3323(3)	2.303(12)	20.1(7)	5.8(3)	3.009(3)
5	0.5216(2)	1.577(4)	13.8(5)	3.7(2)	1.917(1)
6	1.010 07(10)	1.77(1)	15.5(5)	1.9(1)	0.990(1)

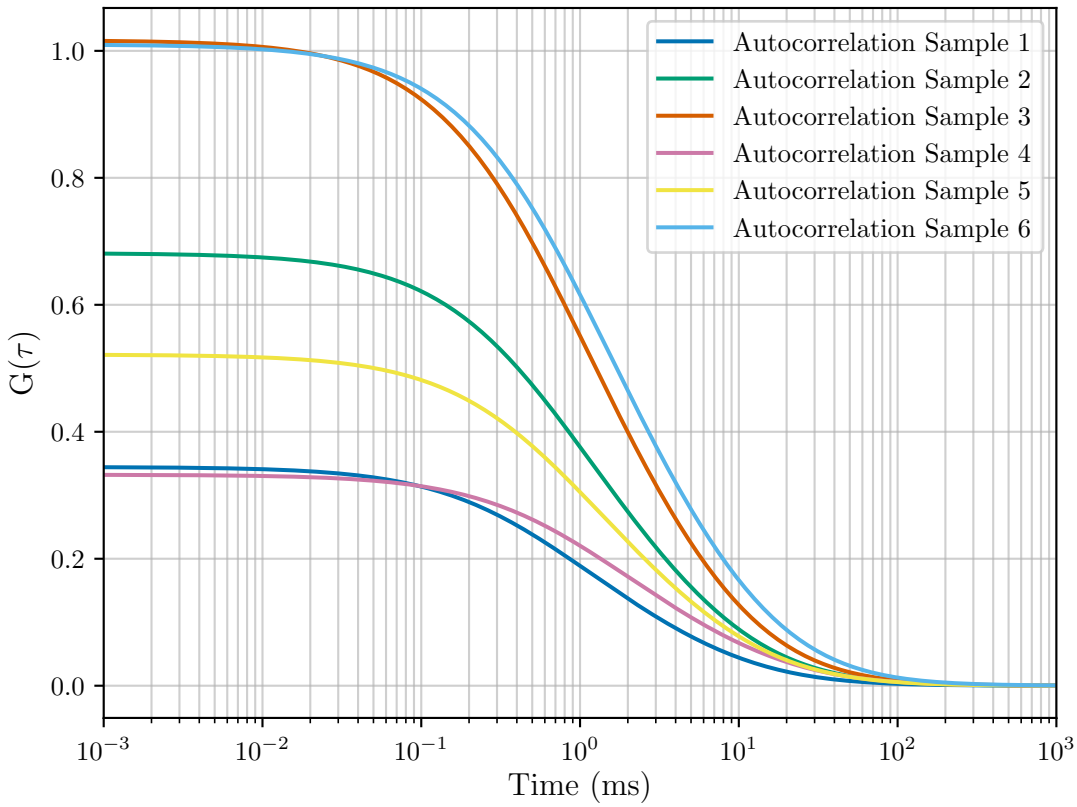


Figure 2.6: All Autocorrelations for the different samples.

with the viscosity at the room temperature (T = 20.0(5) °C) of $\eta = 938(12)$ µPa s. The concentration c in nanoparticles per liter and the average number of particles $\langle N \rangle$ in the lense focus can be derived from the following relations:

$$c = \frac{1}{G_0 \cdot V_{eff}} \tag{2.12}$$

$$\langle N \rangle = \frac{1}{G_0} \tag{2.13}$$

All the calculated values for the different samples and listed in table 2.3. The concentration was converted into the unit nmol L^{-1}, which is also referred to as nano-Molar.
Based on the similar hydrodynamic radius R_H the samples one, two and three might contain the same nanoparticles in different concentrations. The same assumption can also be made for samples five and six.

Bibliography

[1] I. für Angewandte Physik. A student's guide to the fluorescence correlation spectroscopy (fcs) laboratory course.

[2] A. Loman. *Molecular Sizing using Fluorescence Correlation Spectroscopy*. Phd thesis, Division of Mathematics and Natural Sciences of the Georg-August-Universität Göttingen, Göttingen, Germany, 2010. Available at `https://d-nb.info/1043594981/34`.